COL761 Assignment 2: Route Blocking for Reducing Forest Fire Spread

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Task 1: Optimization Formulation

The route blocking problem can be formulated as a combinatorial optimization problem on the directed forest graph $G = (V, E, p)$.

Decision Variables: Let $x_e \in \{0, 1\}$ be a binary decision variable for each route $e \in E$, defined as:

$$x_e = \begin{cases} 1, & \text{if route } e \text{ is blocked} \\ 0, & \text{otherwise} \end{cases}$$

The set of blocked routes is then $R = \{e \in E \mid x_e = 1\}$.

Objective Function: The objective is to minimize the expected total number of burned regions at termination, denoted by $\sigma(R) = \mathbb{E}[|A_\infty|]$.

Constraints: The constraints for this problem are:

1. The total number of blocked routes must exactly equal the budget k:

$$\sum_{e \in E} x_e = k$$

2. The decision variables must be binary:

$$x_e \in \{0, 1\}, \quad \forall e \in E$$

Formulation

Combining the objective and the constraints, the complete optimization problem is formulated as follows:

$$\begin{aligned} \min_x \quad & \sigma(R) \\ \text{subject to} \quad & \sum_{e \in E} x_e = k \\ & x_e \in \{0, 1\}, \quad \forall e \in E \end{aligned}$$

Task 2: NP-Hardness

Theorem: The route blocking problem is NP-hard.

Proof. A general problem is at least as hard as its special cases. Therefore, if we can prove the route blocking problem is NP-hard when all propagation probabilities are restricted to exactly $p = 1$ (deterministic spread), the general stochastic Independent Cascade model is inherently NP-hard by extension. We demonstrate this via a polynomial-time reduction from the **Minimum K-Coverage** problem.

1. Minimum K-Coverage Problem Definition

- **Input:** A universe of elements $U = \{e_1, e_2, \ldots, e_n\}$, a collection of subsets $\mathcal{S} = \{S_1, S_2, \ldots, S_m\}$ where $S_i \subseteq U$, and an integer q.
- **Objective:** Select a subset of subsets $\mathcal{Q} \subseteq \mathcal{S}$ such that $|\mathcal{Q}| = q$, and the number of unique elements covered by their union, $|\bigcup_{S_i \in \mathcal{Q}} S_i|$, is minimized.

2. Graph Construction Given an instance of Minimum K-Coverage, we construct a directed fire-spread graph $G = (V, E, p)$:

- **Nodes:**
  - Create a seed node s (representing A_0).
  - Create a subset node v_{S_i} for each $S_i \in \mathcal{S}$.
  - Create an element node v_{e_j} for each element $e_j \in U$.
  - Create W dummy nodes for each element node v_{e_j} to act as a massive penalty, where $W > m + n$.
- **Edges:**
  - Add directed edges from the seed to subsets: $E_{seed} = \{(s, v_{S_i}) \mid i \in \{1, \ldots, m\}\}$.
  - Add directed edges (v_{S_i}, v_{e_j}) if $e_j \in S_i$.
  - Add directed edges from each v_{e_j} to its respective W dummy nodes.
- **Parameters:** Set propagation probabilities $p = 1.0$ for all edges. Set the route-blocking budget to $k = m - q$.

3. Small Example & Equivalent Graph Consider a Minimum K-Coverage instance where $U = \{1, 2, 3\}$, $\mathcal{S} = \{S_1 = \{1, 2\}, S_2 = \{2, 3\}, S_3 = \{3\}\}$, and we must select $q = 1$ subset. The optimal choice is S_3 (covering 1 element). In our mapped graph, $m = 3$ and $n = 3$, so the budget is $k = 3 - 1 = 2$ edges, and the penalty weight must be $W > 6$ (e.g., $W \geq 7$).

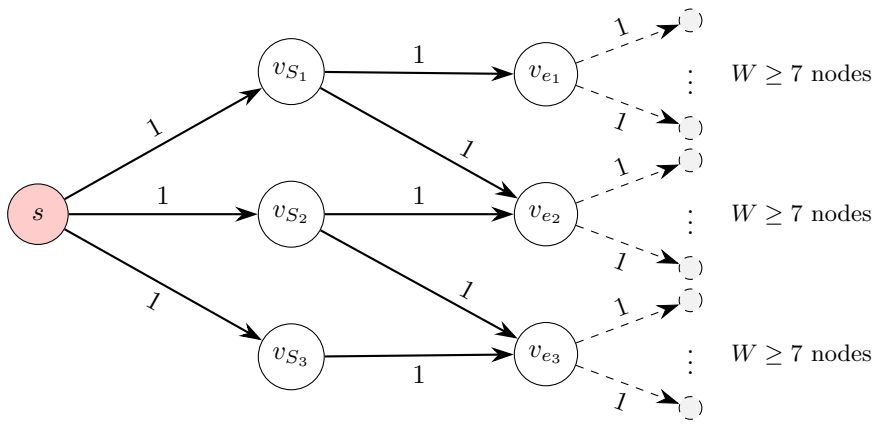

Figure 1: Mapped Influence Blocking graph for the K-Coverage example.

4. Formal Proof of Edge Selection **Lemma:** To minimize the spread, the k chosen edges to be removed must be exclusively of the form (s, v_{S_i}).

Assume for contradiction that the optimal removal set R^* contains an edge $e' \notin E_{seed}$. By graph construction, e' must be strictly downstream of some specific subset node v_{S_i}. We define a replacement edge $e^* \in E_{seed}$ as follows: if $(s, v_{S_i}) \notin R^*$, let $e^* = (s, v_{S_i})$; otherwise, choose any unblocked $(s, v_{S_x}) \notin R^*$ (which is guaranteed to exist since $k < m$).

Consider the new set $R' = (R^* \setminus \{e'\}) \cup \{e^*\}$. By swapping e' with e^*, we strictly increase the number of protected nodes:

- If $e^* = (s, v_{S_i})$, blocking e^* intercepts all paths that e' would have blocked, protecting an equal number of downstream nodes, while additionally protecting at least the node v_{S_i} itself.
- If $e^* = (s, v_{S_x})$, it implies v_{S_i} (and thus e') was already completely isolated from the seed. Thus, e' protected 0 nodes, whereas e^* protects at least the node v_{S_x}.

In either scenario, swapping e' for an edge in E_{seed} strictly reduces the expected spread, yielding $\sigma(R') < \sigma(R^*)$. This mathematically contradicts the assumption that R^* is optimal.

5. Equivalence Argument By the lemma, the optimal strategy strictly allocates its budget to remove k edges from E_{seed}. This effectively disconnects k subsets, leaving exactly $m - k = q$ subset nodes connected to the seed s. Let $\mathcal{Q} \subset \mathcal{S}$ be the set of these q remaining subsets.

Because the spread is deterministic ($p = 1$), the fire spreads from s to the q subset nodes, which then deterministically ignite the element nodes they cover, denoted as $U_{\mathcal{Q}} = \bigcup_{S_i \in \mathcal{Q}} S_i$. Each ignited element node triggers its massive W dummy nodes. The expected spread becomes:

$$\sigma(R) = 1 + q + |U_{\mathcal{Q}}| + W \cdot |U_{\mathcal{Q}}|$$

Since the seed (1), the unblocked subsets (q), and the penalty weight (W) are all fixed constants, minimizing the objective function $\sigma(R)$ is strictly equivalent to minimizing $|U_{\mathcal{Q}}|$. Therefore, choosing k edges to maximize blocking is perfectly equivalent to choosing q subsets such that their union coverage is minimized. Since Minimum K-Coverage is NP-hard, the Route Blocking problem is NP-hard. □

Task 3: Structural Properties

Let $\sigma(R)$ denote the expected fire spread after blocking routes $R \subseteq E$.

1. Monotonicity

Definition: A set function $f : 2^E \to \mathbb{R}$ is monotonically decreasing if for all subsets A, B such that $A \subseteq B \subseteq E$, it holds that $f(A) \geq f(B)$.

Theorem: $\sigma(R)$ is monotonically decreasing with respect to R.

Proof. Assume for contradiction that there exist route sets $R_1 \subseteq R_2 \subset E$ such that $\sigma(R_1) < \sigma(R_2)$.

Let $E_1 = E \setminus R_1$ and $E_2 = E \setminus R_2$ represent the available edges for fire propagation.

$$R_1 \subseteq R_2 \implies E_2 \subseteq E_1$$

For a fixed stochastic realization ω of the edge ignition probabilities, let $A^\omega_\infty(R)$ denote the final set of burned regions when routes R are blocked.

If a directed path exists from the seed set A_0 to a node v using only edges in E_2 under realization ω, that exact same path must also exist in E_1 under ω, since $E_2 \subseteq E_1$.

$$\forall v \in V, \forall \omega : (v \in A^\omega_\infty(R_2) \implies v \in A^\omega_\infty(R_1))$$

$$\implies A^\omega_\infty(R_2) \subseteq A^\omega_\infty(R_1)$$

$$\implies |A^\omega_\infty(R_2)| \leq |A^\omega_\infty(R_1)|$$

Taking the expected value over all possible realizations Ω:

$$\mathbb{E}_{\omega \sim \Omega}[|A^\omega_\infty(R_2)|] \leq \mathbb{E}_{\omega \sim \Omega}[|A^\omega_\infty(R_1)|]$$

$$\implies \sigma(R_2) \leq \sigma(R_1)$$

This mathematically contradicts the initial assumption that $\sigma(R_1) < \sigma(R_2)$. Therefore, $\sigma(R)$ is monotonically decreasing. □

2. Submodularity

Definition: A set function $f : 2^E \to \mathbb{R}$ is submodular if for all $S \subseteq T \subset E$ and $e \in E \backslash T$, the diminishing returns property holds:

$$f(S \cup \{e\}) - f(S) \geq f(T \cup \{e\}) - f(T)$$

Theorem: $\sigma(R)$ is **not** submodular.

Proof. Note that the definition of submodularity evaluates strict algebraic values without absolute functions.

Counterexample Consider the directed path graph G below with initial burning region $A_0 = \{s\}$ and propagation probability $p = 1.0$ on all edges.

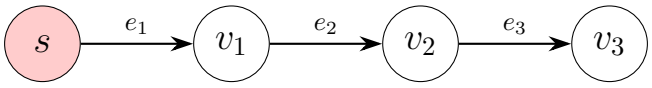

Figure 2: Counterexample path graph for Submodularity.

Let $S = \emptyset$ and $T = \{e_2\}$, satisfying $S \subseteq T$. Let the new edge to be blocked be $e = e_1$.

- **Evaluating marginal change for S:** When $S = \emptyset$, the fire spreads deterministically across all edges. Nodes s, v_1, v_2, v_3 burn, so $\sigma(S) = 4$. Blocking e_1 disconnects the path. Only s burns, so $\sigma(S \cup \{e_1\}) = 1$.
$$\sigma(S \cup \{e_1\}) - \sigma(S) = 1 - 4 = -3$$
- **Evaluating marginal change for T:** When $T = \{e_2\}$ is blocked, the fire propagates along e_1 but stops at v_1. Nodes s, v_1 burn, so $\sigma(T) = 2$. Adding e_1 blocks $\{e_1, e_2\}$. Only s burns, so $\sigma(T \cup \{e_1\}) = 1$.
$$\sigma(T \cup \{e_1\}) - \sigma(T) = 1 - 2 = -1$$

Substituting these marginal changes into the submodularity definition yields:

$$-3 \geq -1$$

This is a mathematical contradiction since -3 is algebraically less than -1. Because the inequality fails, $\sigma(R)$ violates the diminishing returns property and is therefore not submodular. □

Task 4: Algorithm Design

4.1 Final Algorithm

We use a scalable two-stage hybrid Monte-Carlo greedy heuristic:

- **Reachability filtering:** ignore currently unblocked edges whose tail is unreachable from seeds (or beyond hop limit).
- **Stage-1 coarse scoring:** run multiple IC simulations and assign credit backward from burned nodes to successful igniting edges.
- **Stage-2 fine reranking:** shortlist top-L edges from Stage-1, sample common random worlds, and pick edge minimizing average spread.
- **Timeout safety:** write output after each selected edge and also at periodic checkpoints.

4.2 Pseudocode

```
Input: G=(V,E,p), seeds S, budget k, n_random_instances r, hops h
R <- empty
write_output(R)
for t = 1..k:
    C <- unblocked and currently reachable candidate edges
    if C empty: pick any remaining edge, add to R, write_output(R), continue

    // Stage-1: coarse Monte-Carlo credit scoring
    score[e] <- 0 for e in C
    run T_c simulations of IC process on current graph:
        log successful parent edges for each newly burned node
        back-propagate node credit to igniting edges
        accumulate score[e]
    L <- top shortlist_size edges by score

    // Stage-2: fine reranking using common sampled worlds
    for each e in L: avgSpread[e] <- 0
    run T_f sampled live-edge worlds:
        for each e in L:
            compute deterministic spread if e is additionally blocked
            accumulate avgSpread[e]
    choose e* = argmin avgSpread[e]
    R <- R U {e*}
    write_output(R)

if |R| < k: fill remaining with arbitrary unblocked edges, writing each step
return first k edges in R
```

4.3 Complexity

Let $n = |V|$, $m = |E|$, budget k, coarse simulations/round T_c, fine worlds/round T_f, shortlist size L (constant in implementation, max 24).

Per round cost: $O(n+m)$ (filtering) + $O(T_c(n+m))$ (coarse MC) + $O(T_f L(n+m))$ (fine rerank).

Total Time Complexity: $O(k(n+m)(1+T_c+T_f L))$.

Memory is $O(n+m)$ with reusable buffers.

4.4 Approximation Guarantee Discussion

Because σ is not submodular (as proven in Task 3), our implemented solver is not an exact global optimization of σ; it uses Monte-Carlo estimation, proxy scoring, and shortlisting. Therefore, this implementation provides no formal constant-factor approximation guarantee. Furthermore, the standard $(1-1/e)$ theorem applies to the maximization of monotone submodular objectives, and thus cannot be applied to this cardinality-constrained minimization setting with a non-submodular objective.

Task 5: Competitive Evaluation

5.1 Setup

Evaluation used the provided scripts: `A2/q2/Eval/evaluate.py` and `evaluate.sh`. Public settings:

- **Dataset 1:** $k = 50$, $hops = -1$, num_sim = 50
- **Dataset 2:** $k = 20$, $hops = 3$, num_sim = 50

5.2 Results (official num_sim=50)

The following table shows the evaluation results. Final submitted output quality is balanced across both models and robust in runtime.

Method	**Dataset 1** $\rho(R)$	**Dataset 2** $\rho(R)$
Final submitted (current forest_fire.cpp)	0.834620	0.880727

5.3 Stability check (num_sim=500)

Method	**Dataset 1** $\rho(R)$	**Dataset 2** $\rho(R)$
Final submitted	0.779846	0.884162